

Random Class

The `Random` class in Java is part of the `java.util` package and provides methods to generate random numbers of different types, such as `int`, `long`, `double`, and `boolean`. It is commonly used to simulate random events in applications such as games, simulations, and security features.

Key Features of the Random Class:

- The `Random` class can generate random numbers for different data types.
 - The randomness is pseudo-random, meaning the numbers are generated based on an initial seed value.
 - You can create an instance of `Random` and call its methods to generate random values.
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1. Importing the Random Class

Before using the `Random` class, you need to import it:

```
import java.util.Random;
```

2. Creating an Instance of Random

You can create an instance of the `Random` class using the constructor. You can either use the default constructor (which uses the current time as the seed) or provide a custom seed value.

Syntax:

```
Random rand = new Random();           // Default constructor
Random randWithSeed = new Random(seed); // Constructor with a seed value
```

3. Generating Random Numbers

Generate a Random Integer

The `nextInt()` method generates a random integer. You can specify a bound to limit the range of random numbers.

Syntax:

```
int randInt = rand.nextInt(); // Generates any integer
int randIntInRange = rand.nextInt(100); // Generates an integer between 0
and 99
```

Example:

```
import java.util.Random;

public class RandomExample {
    public static void main(String[] args) {
        Random rand = new Random();

        // Generating a random integer
        int randInt = rand.nextInt(); // Random integer
        System.out.println("Random integer: " + randInt);

        // Generating a random integer between 0 and 99
        int randIntInRange = rand.nextInt(100); // Generates a number
between 0 and 99
        System.out.println("Random integer between 0 and 99: " +
randIntInRange);
    }
}
```

Generate a Random Boolean

The `nextBoolean()` method returns a random boolean value (`true` or `false`).

Syntax:

```
boolean randBool = rand.nextBoolean();
```

Example:

```
import java.util.Random;

public class RandomExample {
    public static void main(String[] args) {
        Random rand = new Random();
```

```
        // Generating a random boolean
        boolean randBool = rand.nextBoolean();
        System.out.println("Random boolean: " + randBool);
    }
}
```

Generate a Random Double

The `nextDouble()` method generates a random number between 0.0 (inclusive) and 1.0 (exclusive).

Syntax:

```
double randDouble = rand.nextDouble();
```

Example:

```
import java.util.Random;

public class RandomExample {
    public static void main(String[] args) {
        Random rand = new Random();

        // Generating a random double
        double randDouble = rand.nextDouble();
        System.out.println("Random double between 0.0 and 1.0: " +
randDouble);
    }
}
```

Generate a Random Long

The `nextLong()` method generates a random long value.

Syntax:

```
long randLong = rand.nextLong();
```

Example:

```
import java.util.Random;

public class RandomExample {
    public static void main(String[] args) {
        Random rand = new Random();

        // Generating a random long
        long randLong = rand.nextLong();
        System.out.println("Random long: " + randLong);
    }
}
```

The `Random` class is a powerful tool in Java for generating random numbers, which can be used for a wide range of applications, including simulations, games, and security. It provides multiple methods to generate different types of random values, such as integers, booleans, doubles, and longs.

Key methods include:

- `nextInt()`
- `nextDouble()`
- `nextLong()`
- `nextBoolean()`
- `nextFloat()`